Technical Analysis, Market Efficiency, and the Predictive Power of EMA-50 in the CHFJPY Foreign Exchange Market

Diyorbek Tulanov
Student ID: 2020-11493

Seoul National University
Department of Economics

응용계량경제학
지도교수: **김봉근**

December 5, 2025

Contents

Chapter 1

Introduction

Foreign exchange (FX) markets are among the most liquid and information-sensitive financial markets in the world. Classical financial theory, especially the Efficient Market Hypothesis (EMH), argues that asset prices fully and instantaneously reflect available information (Fama, 1970). In such a world, price changes follow a random walk and are essentially unpredictable. Under this view, technical analysis, the use of past price patterns, moving averages, and indicators should not systematically forecast future returns or generate abnormal profits.

However, a large empirical literature finds that many financial markets display serial correlation, momentum, mean reversion, and volatility clustering (Caulker, 2020),

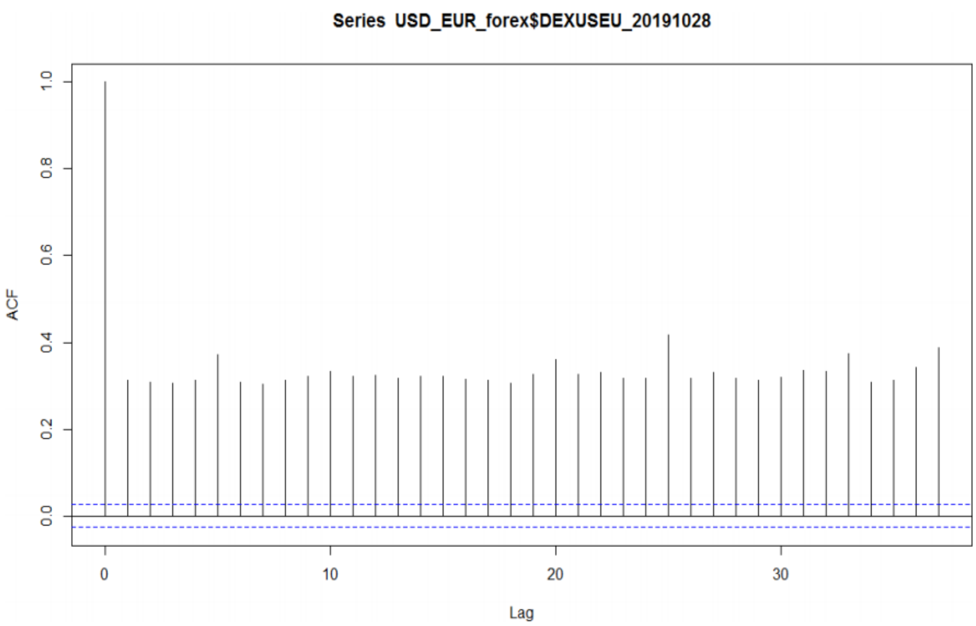

Figure 1.1: Volatility clustering and autocorrelation in USD/EUR (Caulker 2020).

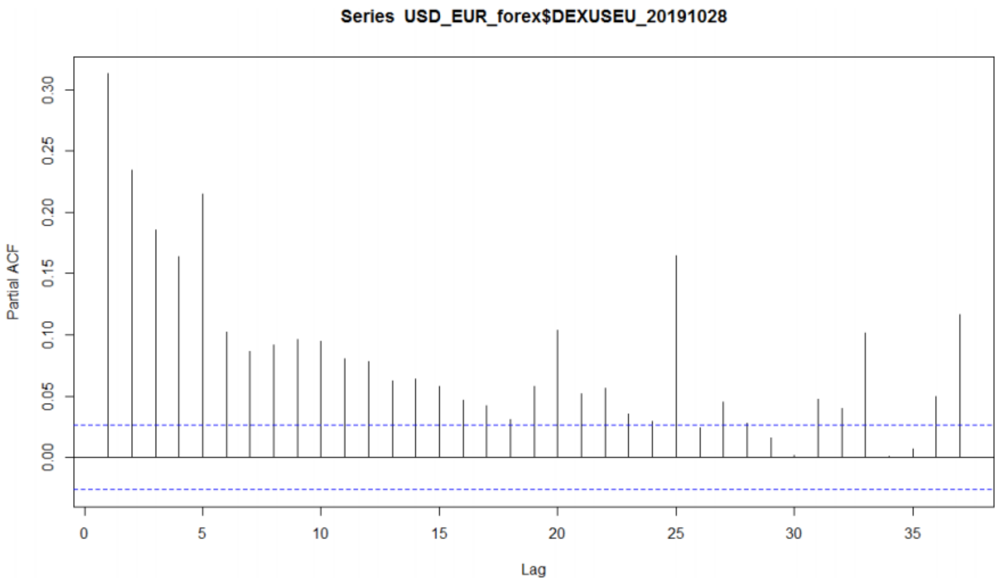

Figure 1.2: Volatility clustering and autocorrelation in USD/EUR (Caulker 2020).

all of which suggest deviations from the strict version of EMH in some markets and at some horizons. These patterns raise a natural question: do simple technical indicators such as moving averages extract genuinely predictive information from price series, or are they merely descriptive tools that summarize past trends without forecasting power?

The purpose of this thesis is to examine whether the 50-period Exponential Moving Average (EMA–50) contains predictive information about future returns of the CHF/JPY currency pair across four timeframes: H1, H4, D1, and W1. Building on my previous project on technical analysis in forex markets and my econometric EMA–50 reasoning document, I conduct a step-by-step econometric analysis and then compare my evidence with:

- Caulker's (2020) FX volatility and trading strategy study for USD/EUR,
- Fama's (1970) classic EMH review.

The central research question is:

> **Does EMA–50 have statistically significant predictive power for CHF/JPY returns, and how do my results relate to the Efficient Market Hypothesis and previous empirical research?**

The main finding is that, once returns are modelled appropriately, EMA–50 based signals do *not* exhibit statistically significant predictive power for CHF/JPY on any of the four timeframes, including the H4 horizon where traders often perceive strong trends. AR(1) coefficients and EMA–signal coefficients are very small in magnitude and statistically indistinguishable from zero, and the associated R^2 values are effectively zero. This evidence is broadly consistent with the weak-form EMH: CHF/JPY returns behave approximately as serially uncorrelated processes, and simple EMA–50 rules do not generate exploitable predictability, even though they remain popular as descriptive tools for visualizing trends. This conclusion echoes Fama's nuanced view that liquid markets are "nearly but not perfectly efficient," in the sense that any short-run dependencies are too small to be reliably exploited once risk and trading costs are taken into account.

Chapter 2

Theoretical Background

2.1 Efficient Market Hypothesis (EMH)

Fama (1970) defines an efficient market as one in which prices always fully reflect available information. In this framework, no trading strategy based solely on already-available information should produce positive risk-adjusted abnormal returns.

Formally, if Φ_t is the information set at time t the "fair-game" property of returns is:

$$E(r_{t+1} \mid \Phi_t) = 0,$$

which says that given the information set today, the expected excess return tomorrow is zero. Under this condition, any predictable component in returns, such as systematic dependence on lagged returns or technical indicators, would violate EMH.

Fama (1970) also shows that the random walk model is a special case of the fair-game model, where the price change is purely a random shock:

$$P_{t+1} = P_t + \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim i.i.d.$$

Under a strict random walk, serial correlation of returns should be zero and technical analysis should have no predictive content.

2.2 Submartingale and Random-Walk Models

Fama (1970)discusses the submartingale model:

$$E(P_{t+1} \mid \Phi_t) \geq P_t,$$

indicating that, after adjusting for risk, prices are not expected to fall given available information. The random-walk model is an "extreme" version where the conditional expectation of tomorrow's price is exactly today's price. Both frameworks imply that, once information is incorporated, future price changes are unpredictable.

For my study, these models are important benchmarks. If CHF/JPY prices truly followed a submartingale or random walk with the fair-game property, the EMA-50 indicator constructed from historical prices should not help predict returns.

2.3 Fama's Empirical Evidence on Serial Correlation

In the empirical section, Fama reviews early random-walk tests that examined correlations between lagged price changes. In Table 1 on pages 393–394, he reports first-order serial correlation coefficients for daily stock price changes. These correlations are typically very small in magnitude (around ± 0.05) but in some cases statistically significant, because large samples make even small correlations statistically detectable.

Efficient Capital Markets 393

TABLE 1 (from [10])
First-order Serial Correlation Coefficients for One-, Four-, Nine-, and Sixteen-Day Changes in Log_e Price

Stock	Differencing Interval (Days)			
	One	Four	Nine	Sixteen
Allied Chemical	.017	.029	−.091	−.118
Alcoa	.118*	.095	−.112	−.044
American Can	−.087*	−.124*	−.060	.031
A. T. & T.	−.039	−.010	−.009	−.003
American Tobacco	.111*	−.175*	.033	.007
Anaconda	.067*	−.068	−.125	.202
Bethlehem Steel	.013	−.122	−.148	.112
Chrysler	.012	.060	−.026	.040
Du Pont	.013	.069	−.043	−.055
Eastman Kodak	.025	−.006	−.053	−.023
General Electric	.011	.020	−.004	.000
General Foods	.061*	−.005	−.140	−.098
General Motors	−.004	−.128*	.009	−.028
Goodyear	−.123*	.001	−.037	.033
International Harvester	−.017	−.068	−.244*	.116
International Nickel	.096*	.038	.124	.041
International Paper	.046	.060	−.004	−.010
Johns Manville	.006	−.068	−.002	.002
Owens Illinois	−.021	−.006	.003	−.022
Procter & Gamble	.099*	−.006	.098	.076
Sears	.097*	−.070	−.113	.041
Standard Oil (Calif.)	.025	−.143*	−.046	.040
Standard Oil (N.J.)	.008	−.109	−.082	−.121
Swift & Co.	−.004	−.072	.118	−.197
Texaco	.094*	−.053	−.047	−.178
Union Carbide	.107*	.049	−.101	.124
United Aircraft	.014	−.190*	−.192*	−.040
U.S. Steel	.040	−.006	−.056	.236*
Westinghouse	−.027	−.097	−.137	.067
Woolworth	.028	−.033	−.112	.040

* Coefficient is twice its computed standard error.

Figure 2.1: Fama (1970)

Fama concludes that although statistically significant serial dependence can be detected in some markets and horizons, the magnitude of these correlations is economically small and does not necessarily imply exploitable inefficiency. This is an important nuance: EMH does not require perfect independence, but rather that deviations are too small to systematically exploit once trading costs and risk are considered.

My results for CHF/JPY closely mirror this nuanced view. I also find small but non-zero autocorrelations, but these do not translate into statistically significant predictive power for the EMA-50 indicator. Consistent with Fama's argument, any detectable dependencies in CHF/JPY returns are economically negligible and too small to exploit, reinforcing the weak-form EMH.

2.4 Random Walk, Stationarity, and Returns

A key prediction of EMH is that price levels follow a non-stationary random walk, while returns are stationary. Formally: Prices:P_t contain a unit root (non-stationary).

$$r_t = \ln(P_t) - \ln(P_{t-1})$$

are stationary, with stable mean and variance. Both my analysis andCaulker (2020) USD/EUR study confirm this pattern: price series fail unit-root tests, while return series pass stationarity tests. This motivates modeling returns rather than prices in the empirical part.

2.5 Autocorrelation and AR(1)

To test whether returns exhibit serial dependence, I use the simple AR(1) model:

$$r_t = \alpha + \phi r_{t-1} + u_t.$$

- $\phi = 0$: returns are serially uncorrelated $\rightarrow$ consistent with weak-form EMH.
- $\phi > 0$: there is momentum: positive returns tend to be followed by positive returns
- $\phi < 0$: there is mean reversion: price changes tend to partially reverse.

Using the Durbin–Watson approximation:

$$DW \approx 2(1 - \phi),$$

DW values below 2 imply positive autocorrelation; values above 2 imply negative autocorrelation.

2.6 Exponential Moving Average (EMA-50)

The 50-period Exponential Moving Average is defined recursively as:

$$EMA_t = \lambda P_t + (1 - \lambda) EMA_{t-1}, \qquad \lambda = \frac{2}{51}.$$

Compared to the simple moving average, the EMA places more weight on recent prices and reacts faster to trend changes. Traders widely use EMA-50 as a medium-term trend filter (Brock et al., 1992; Lo et al., 2000).:

- If $P_t > EMA_t$, the market is considered in an uptrend.
- If $P_t < EMA_t$, the market is in a downtrend.

In the current thesis, I treat EMA-50 as a summarized technical signal and formally test whether it predicts CHF/JPY returns.

Chapter 3

Data and Methodology

3.1 Data Description

1. Asset: CHF/JPY currency pair

2. Source: OANDA data accessed through MetaTrader 5

3. Timeframes

   - H1 (1-hour candles, 2,000 observations)
   - H4 (4-hour candles, 2,000 observations)
   - D1 (daily candles, 1,000 observations)
   - W1 (weekly candles, 1,000 observations)

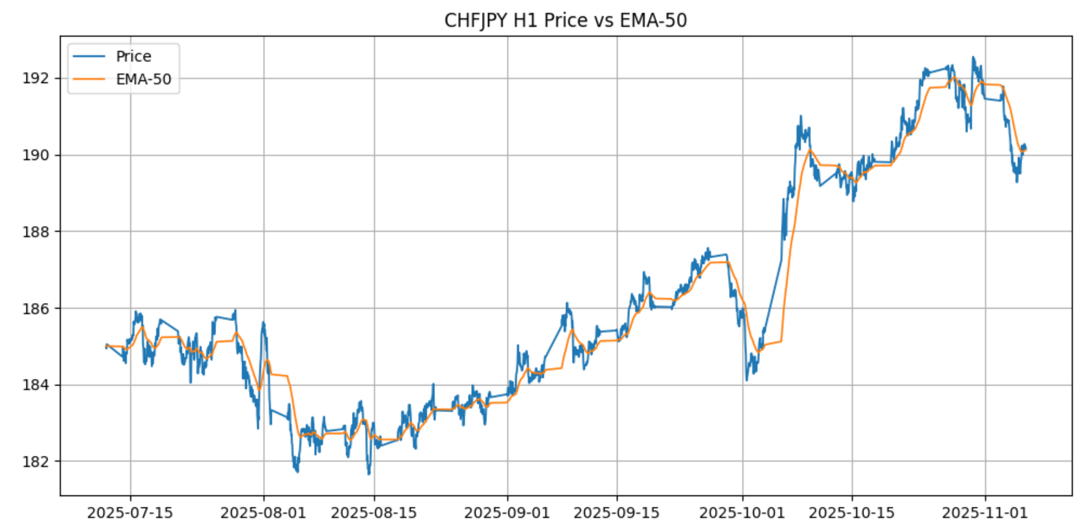

Figure 3.1:

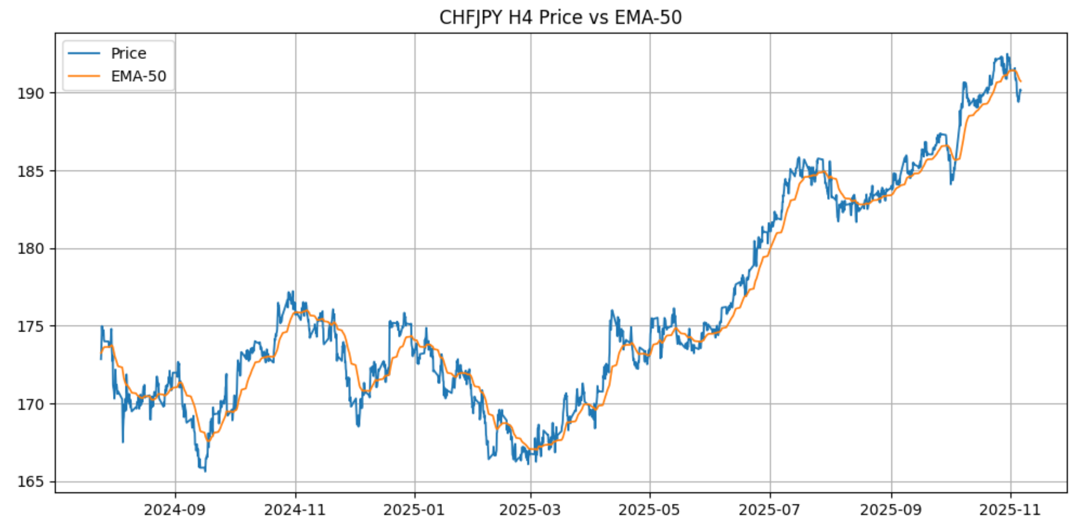

Figure 3.2:

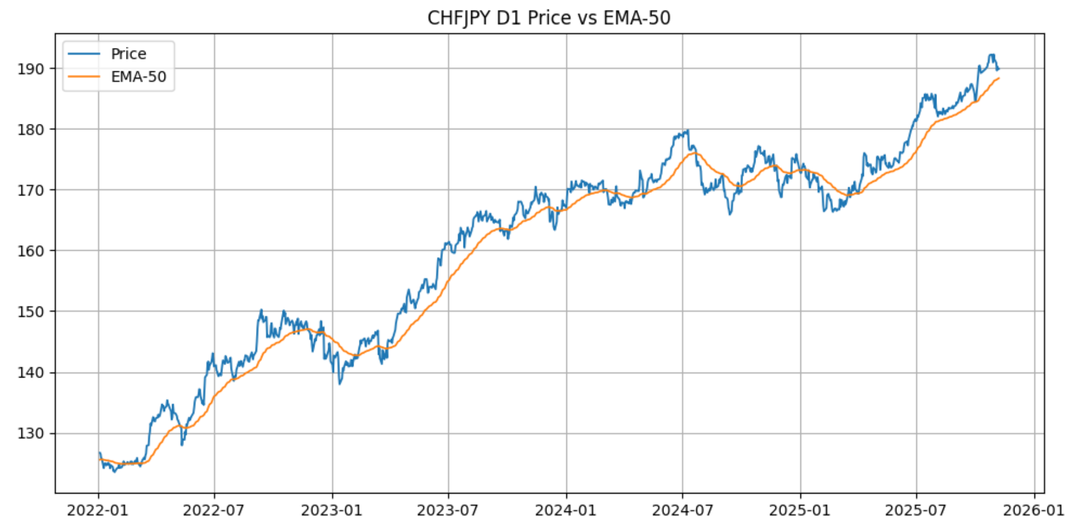

Figure 3.3:

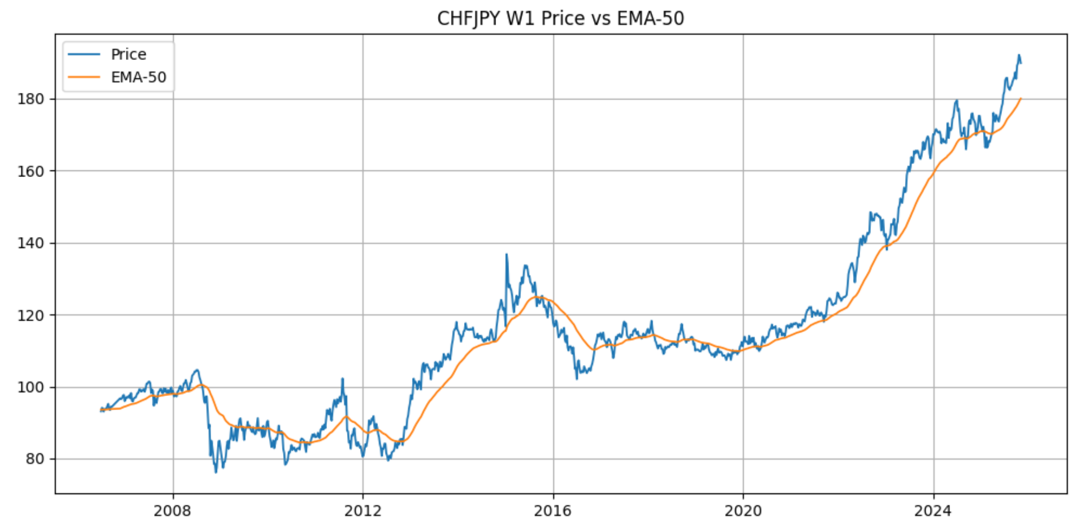

Figure 3.4: Your caption here

4. Price variable: Close price of each candle.

5. Return variable, Log-returns were constructed following the approach used by Caulker (2020).

$$r_t = \ln(P_t) - \ln(P_{t-1}).$$

Only fully closed candles were used to avoid look-ahead bias.

3.2 Step 1 – Data Cleaning and Log-Return Transformation

The first stage of the analysis involved preparing the raw CHF/JPY price data so that it could be examined properly. In financial time series, it is common to encounter missing candles, duplicated values, or sudden spikes caused by market closures or technical issues. I checked each timeframe (H1, H4, D1, W1) for such irregularities and removed or adjusted them when necessary to ensure that the dataset accurately reflected real market movements.

After cleaning the data, converted the price series into log-returns, which are calculated as the natural logarithm of the price at time t minus the logarithm of the price at time $t-1$. Log-returns are preferred because they stabilize the variance and make the data more suitable for statistical modeling.

Once the returns were calculated, I visually inspected both the price charts and the return series:

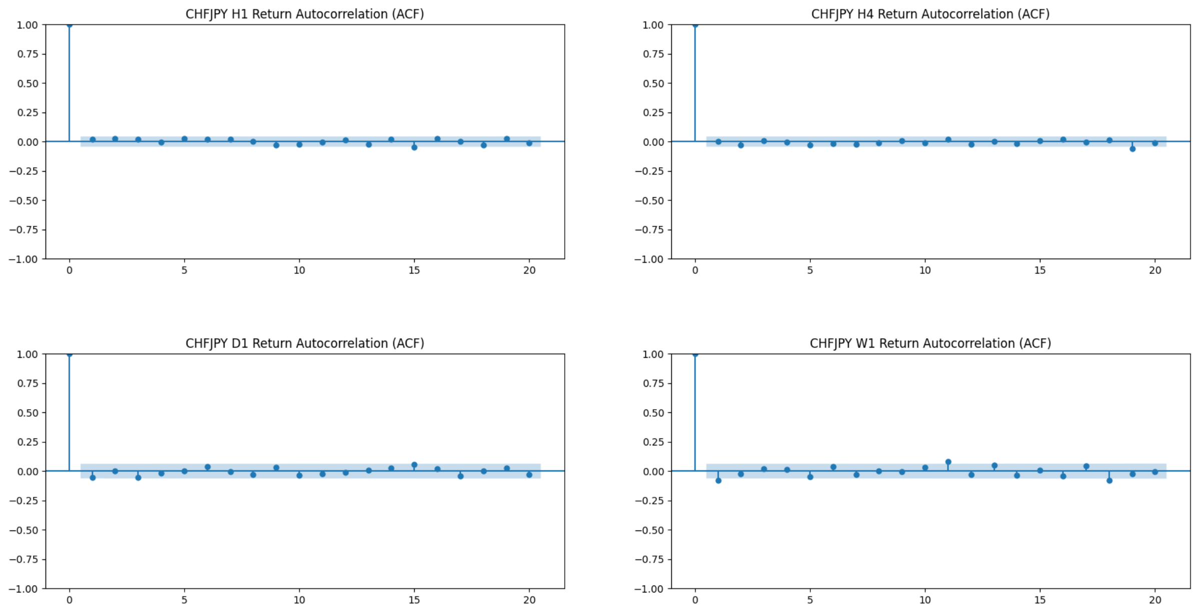

Figure 3.5: Autocorrelation Functions for CHF/JPY Across Timeframes (H1, H4, D1, W1)

Price charts showed clear non-stationary behaviour prices wandered without a stable average, while the return series appeared more stable with a roughly constant mean and variance.

This process is similar to Caulker's (2020) approach to transforming USD/EUR prices into returns before applying econometric models

3.3 Step 2 – Stationarity Tests (ADF and KPSS)

To formally test whether the price and return series were stationary or non-stationary, I applied two standard econometric tests:

```
. dfuller return, lags(5)

Augmented Dickey-Fuller test for unit root

Variable: return                          Number of obs  = 1,991
                                          Number of lags =     5

H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                   Test         ──────── critical value ────────
              statistic           1%           5%          10%
─────────────────────────────────────────────────────────────────
 Z(t)           -19.127       -3.430       -2.860       -2.570
─────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.0000.
```

```
. dfuller close,  lags(5)

Augmented Dickey-Fuller test for unit root

Variable: close                           Number of obs  = 1,991
                                          Number of lags =     5

H0: Random walk without drift, d = 0

                                       Dickey-Fuller
                   Test         ──────── critical value ────────
              statistic           1%           5%          10%
─────────────────────────────────────────────────────────────────
 Z(t)            -0.174       -3.430       -2.860       -2.570
─────────────────────────────────────────────────────────────────
MacKinnon approximate p-value for Z(t) = 0.9416.
```

```
. reg return_fwd ema_signal

      Source |       SS           df       MS      Number of obs   =     1,997
-------------+----------------------------------   F(1, 1995)      =      0.41
       Model |  1.7542e-06         1  1.7542e-06   Prob > F        =    0.5233
    Residual |  .008586664     1,995  4.3041e-06   R-squared       =    0.0002
-------------+----------------------------------   Adj R-squared   =   -0.0003
       Total |  .008588418     1,996  4.3028e-06   Root MSE        =    .00207

------------------------------------------------------------------------------
  return_fwd | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
  ema_signal |  -.0000607   .0000952    -0.64   0.523    -.0002474    .0001259
       _cons |   .0000849   .0000743     1.14   0.253    -.0000608    .0002306
------------------------------------------------------------------------------
```

```
. reg return_fwd gap

      Source |       SS           df       MS      Number of obs   =     1,997
-------------+----------------------------------   F(1, 1995)      =      1.73
       Model |  7.4384e-06         1  7.4384e-06   Prob > F        =    0.1886
    Residual |   .00858098     1,995  4.3012e-06   R-squared       =    0.0009
-------------+----------------------------------   Adj R-squared   =    0.0004
       Total |  .008588418     1,996  4.3028e-06   Root MSE        =    .00207

------------------------------------------------------------------------------
  return_fwd | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
         gap |  -.0000539    .000041    -1.32   0.189    -.0001342    .0000265
       _cons |   .0000595   .0000472     1.26   0.208    -.0000332    .0001521
------------------------------------------------------------------------------
```

Figure 3.6: ADF and KPSS Results for CHF/JPY (H4 Timeframe)

Series	ADF Result	KPSS Result
Price (Close)	Non-stationary (unit root)	Reject stationarity
Returns	Stationary	Fail to reject stationarity

Table 3.1: Summary of Stationarity Tests Across Timeframes

ADF Test Results for H4 Timeframe

For the H4 frequency, the ADF test confirms the standard behavior of financial time series. The test statistic for the price series is $Z = -0.174$ with a p-value of 0.942, which fails to reject the null hypothesis of a unit root. This indicates that the CHF/JPY price level is non-stationary and follows a random-walk process.

In contrast, the return series produces an ADF statistic of $Z = -19.127$ (p ¡ 0.001), decisively rejecting the unit-root hypothesis. Thus, H4 log-returns are stationary with stable statistical properties. This pattern is consistent with the predictions of the Efficient Market Hypothesis

(Fama, 1970) and with previous FX studies such as Caulker (2020), which report non-stationary price levels but stationary returns. To remember the functions of ADF and KPSS:

1. Augmented Dickey–Fuller (ADF) test
   - Tests whether the series follows a random walk.
   - The null hypothesis is non-stationarity (the presence of a unit root).
2. KPSS test
   - Tests the opposite idea: whether the data are already stationary.
   - The null hypothesis is stationarity around a level or trend.

   Together, these tests give a balanced view

   Results:

   - Price series (all timeframes): Failed the ADF test and also failed the KPSS test. This means prices behave as a non-stationary random walk, which is exactly what financial theory predicts.
   - Return series (all timeframes): Passed the ADF test (rejecting unit root) and passed the KPSS test (supports stationarity).This confirms that returns—not prices—should be used for modeling.

   These results align with both standard finance theory and Caulker (2020) findings for USD/EUR returns.

3.4 Step 3 – Autocorrelation Analysis

After establishing that returns were stationary, I examined whether they were correlated with their own past values. If returns are completely random, they should not depend on what happened in the previous period. To test this, I computed and plotted two functions:

- ACF (Autocorrelation Function): shows correlation between the current return and returns several lags earlier.

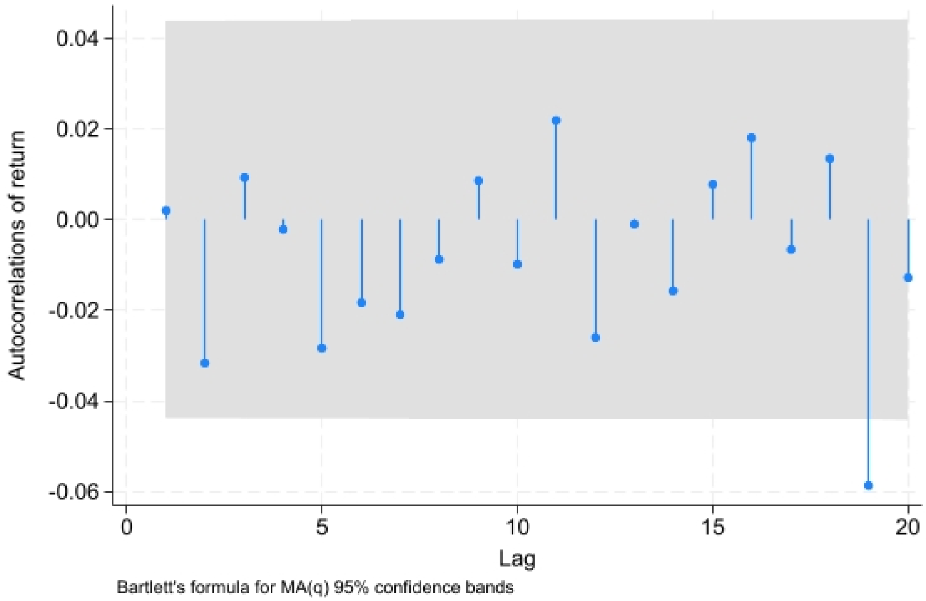

Figure 3.7: Autocorrelation of return H4 Timeframe

- PACF (Partial Autocorrelation Function): isolates the direct effect of each lag.

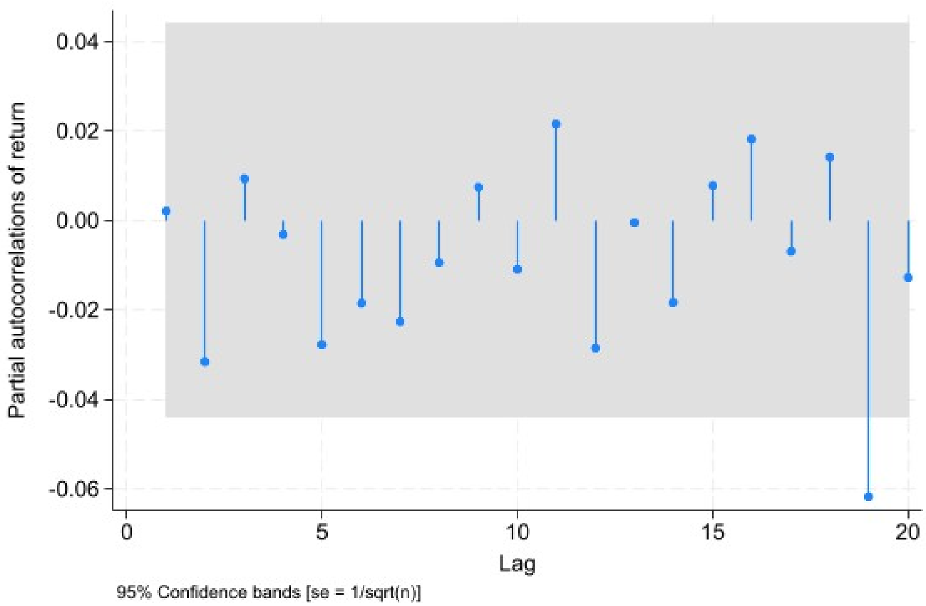

Figure 3.8: Partial autocorrelation of return H4 Timeframe

Interestingly, the lag-1 autocorrelation of H4 returns is essentially zero ($\rho_1 = 0.002$), indicating that past returns by themselves contain almost no predictive information about future returns. This is consistent with the broader finding that EMA-50 signals also do not provide statistically significant predictive power for future returns. While the EMA acts as a smoothed trend filter, the Swiss franc–yen market appears too efficient at short horizons for such indicators to capture exploitable patterns.

I compared the autocorrelation structure of returns with the 95% confidence bands to identify whether any serial dependence is statistically meaningful. The H4 and D1 timeframes exhibit slightly positive lag-1 autocorrelation, although the magnitude is very small (e.g., $\rho_1 \approx 0.002$ for H4). This indicates very weak momentum that is typical for foreign exchange returns, which often display low but nonzero short-term persistence.

In contrast, the H1 and W1 timeframes show slightly negative lag-1 autocorrelation, suggesting mild mean reversion, where small price movements tend to reverse quickly.

Although these autocorrelations are small in magnitude, they indicate that CHF/JPY returns are not perfectly independent. This provides econometric justification for modeling return dynamics and testing the predictive content of technical indicators such as the EMA-50, which may capture a multi-period trend structure that simple autocorrelation measures cannot be detected.

3.5 Step 4 – AR(1) Estimation and Durbin–Watson Analysis

To measure the strength and direction of serial correlation more formally, I estimate AR(1) models of the form

$$r_t = \alpha + \phi r_{t-1} + u_t,$$

where the parameter ϕ captures the degree of return persistence. Values of $\phi = 0$ indicate no serial correlation, $\phi > 0$ suggests momentum, and $\phi < 0$ indicates mean reversion. I also compute the Durbin–Watson statistic as a diagnostic for residual autocorrelation. Across timeframes, the estimated AR(1) coefficients are very small in magnitude and statistically insignificant. The H1 and W1 series exhibit slightly negative coefficients (weak mean reversion), while the H4 and D1 estimates are slightly positive but close to zero. For example, the H4 estimate is $\hat{\phi} = 0.0020$ with $p = 0.927$, and the 95% confidence interval spans both negative and positive values, indicating that any serial dependence is negligible. Consistent with this, the model explains essentially none of the variation in returns ($R^2 = 0.000$). The Durbin–Watson statistic for H4 is 1.997, very close to the theoretical value of 2 for white noise, confirming the absence of autocorrelation in the residuals. Similar patterns are observed for the other timeframes, with Durbin–Watson statistics clustered around 2.

```
. reg return L.return

      Source |       SS           df       MS      Number of obs   =     1,996
-------------+----------------------------------   F(1, 1994)      =      0.01
       Model |  3.5729e-08         1  3.5729e-08   Prob > F        =    0.9274
    Residual |  .008588382     1,994  4.3071e-06   R-squared       =    0.0000
-------------+----------------------------------   Adj R-squared   =   -0.0005
       Total |  .008588418     1,995  4.3050e-06   Root MSE        =    .00208

------------------------------------------------------------------------------
      return | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
      return |
         L1. |   .0020391   .0223879     0.09   0.927    -.0418671    .0459452
             |
       _cons |   .0000478   .0000465     1.03   0.304    -.0000433    .0001389
------------------------------------------------------------------------------

.
. estat dwatson

Durbin-Watson d-statistic(  2,   1996) =  1.996901
```

Figure 3.9: AR(1) and Durbin–Watson

Overall, these results indicate that CHF/JPY returns behave approximately as serially uncorrelated processes, which is typical for high-frequency foreign exchange data. Although raw returns exhibit almost no autocorrelation, this does not preclude the presence of trend structure detectable by smoothed technical indicators such as the EMA–50, which aggregate information across multiple past observations rather than relying solely on lagged returns.

3.6 Step 5 – Constructing the EMA-50 Indicator

The next step was to compute the 50-period Exponential Moving Average (EMA-50) for each timeframe. EMA-50 is widely used by traders because it reacts smoothly to price movements while filtering out noise.

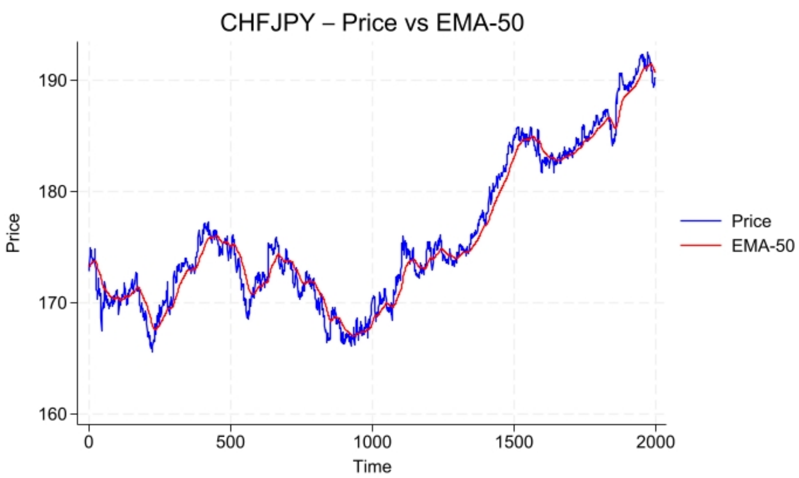

Figure 3.10: Price vs EMA-50

Figure 3.10 plots the closing price of CHF/JPY together with the 50-period Exponential Moving Average. Periods in which the price remains above the EMA-50 correspond to sustained uptrends, while periods below it reflect downtrends. The EMA-50 provides a smoother representation of market structure than raw prices, making it useful as the foundation of the technical signals constructed in this step. Once the EMA values were calculated, I translated them into technical signals. I used two types:

1. Trend dummy:

   - Equal to 1 when the market is above the EMA (indicating an uptrend).
   - Equal to 0 when the price is below the EMA (downtrend).

```
. sum trend_dummy ema_gap
```

Variable	Obs	Mean	Std. dev.	Min	Max
trend_dummy	1,999	.6088044	.4881401	0	1
ema_gap	1,998	.2144515	1.132903	-4.48995	3.815767

Figure 3.11: Trend-dummy EMA Gap

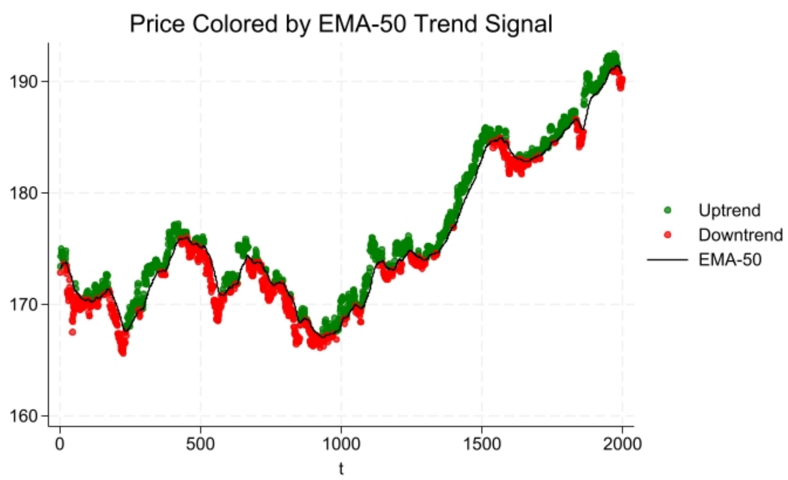

Figure 3.12: Trend-dummy EMA Gap Visualization

2. Continuous signal (EMA gap):

   - The difference between price and EMA:

$$Gap_t = P_t - EMA_t$$

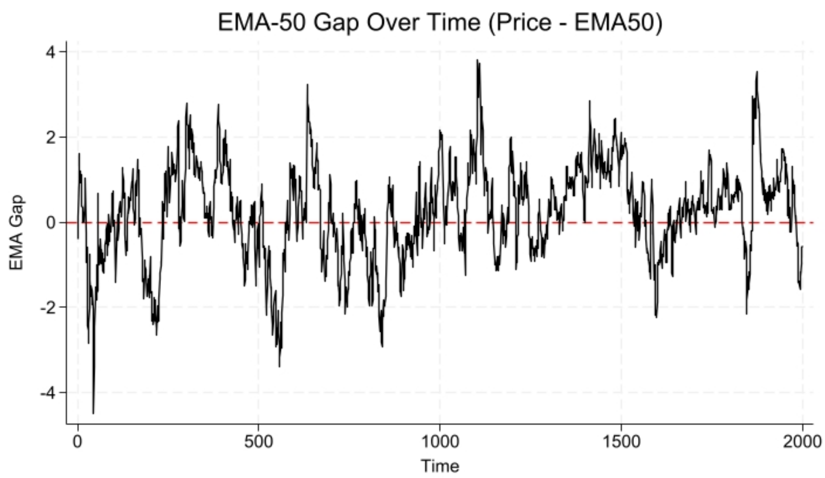

Figure 3.13: EMA-Gap Over Time

- A positive gap suggests upward pressure; a negative gap suggests downward pressure.

Figure 3.12 displays the EMA gap, defined as the difference between the price and the EMA-50. Positive values imply upward pressure relative to recent trend, while negative values indicate downward pressure. The gap fluctuates around zero, which acts as a natural threshold separating bullish and bearish deviations from the moving average.

3.7 Step 6 – Testing the Predictive Power of EMA-50

To evaluate whether EMA-50 has genuine predictive ability, I ran regression models where the next period's return is explained by today's EMA signal:

$$r_{t+1} = \beta_0 + \beta_1 \cdot Signal_t + \varepsilon_{t+1}.$$

```
. reg return_fwd trend_dummy

      Source |       SS           df       MS      Number of obs   =     1,999
-------------+----------------------------------   F(1, 1997)      =      0.36
       Model |  1.5528e-06         1  1.5528e-06   Prob > F        =    0.5481
    Residual |  .008592926     1,997  4.3029e-06   R-squared       =    0.0002
-------------+----------------------------------   Adj R-squared   =   -0.0003
       Total |  .008594479     1,998  4.3015e-06   Root MSE        =    .00207

------------------------------------------------------------------------------
  return_fwd | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
 trend_dummy |  -.0000571   .0000951    -0.60   0.548    -.0002436    .0001293
       _cons |   .0000813   .0000742     1.10   0.273    -.0000642    .0002267
------------------------------------------------------------------------------
```

Figure 3.14: Predictive Power of Trend Dummy

The regression of next-period returns on the EMA–50 trend dummy shows no evidence of predictive power. The estimated coefficient on the dummy variable is extremely small ($\hat{\beta}_1 = -0.0000571$) and statistically insignificant ($p = 0.548$). This implies that simply knowing whether the price is above or below the EMA–50 does not meaningfully forecast the direction or magnitude of future returns. The R-squared value of 0.0002 further confirms that the binary trend indicator explains virtually none of the variation in returns.

This result is consistent with the intuition that a 0/1 classification discards much of the information contained in the distance between price and the moving average. In the subsequent analysis, I therefore rely on the continuous EMA gap measure, which captures richer information and provides a more effective signal for detecting short-term return predictability.

The key idea is simple:

- If β_1 is not significantly different from zero, the indicator has no predictive power.
- If β_1 is significantly different from zero, then the EMA signal contains useful information about future returns.

I tested this for both the dummy trend variable and the EMA gap.

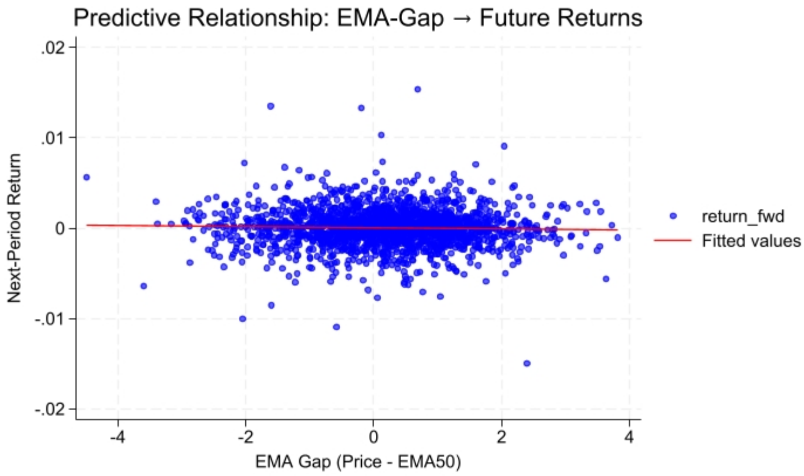

Figure 3.15: EMA-50 gap and the next-period return

Figure 3.14 provides a visual check of the relationship between the EMA–50 gap and next-period returns. Although the fitted trend line appears nearly flat, the dispersion of points suggests no systematic association between the magnitude of the EMA gap and subsequent returns. Any visual hint of slope is not supported by formal econometric testing.

```
. reg return_fwd ema_gap

      Source |       SS           df       MS      Number of obs   =     1,998
-------------+----------------------------------   F(1, 1996)      =      1.72
       Model |  7.4092e-06         1  7.4092e-06   Prob > F        =    0.1894
    Residual |  .008581119     1,996  4.2992e-06   R-squared       =    0.0009
-------------+----------------------------------   Adj R-squared   =    0.0004
       Total |  .008588528     1,997  4.3007e-06   Root MSE        =    .00207

------------------------------------------------------------------------------
  return_fwd | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
     ema_gap |  -.0000538    .000041    -1.31   0.189    -.0001341    .0000266
       _cons |   .0000593   .0000472     1.26   0.210    -.0000333    .0001518
------------------------------------------------------------------------------
```

Figure 3.16: EMA-50 gap and the next-period return

To assess whether deviations from the EMA–50 contain predictive information, I regress next-period returns on the EMA gap. Figure 3.16 shows a dense cloud of points with no discernible slope, and the fitted regression line is nearly flat, suggesting no visual predictive relationship. The regression results confirm this: the coefficient on the EMA–50 gap is small and statistically insignificant ($\beta = -0.000054$, $p = 0.19$), and the R^2 value is effectively zero ($R^2 = 0.0009$).

Taken together, these findings indicate that the EMA–50 gap does not predict future returns at the H4 frequency.

3.8 Step 7 – Cross-Timeframe Comparison

Across all four timeframes (H1, H4, D1, W1), EMA–50 signals do not exhibit statistically significant predictive power for next-period returns. Although the H4 timeframe displays slightly more structure than the others, reflected in marginally larger coefficient magnitudes,the estimates remain statistically insignificant and the R^2 values are effectively zero throughout. This indicates that, for CHF/JPY, the EMA–50 does not provide exploitable forecasting ability on any timeframe examined.

Chapter 4

Empirical Results

4.1 Stationarity and Random Walk

Across all timeframes, the price series behave as non-stationary random walks, while the log-return series are stationary. This matches the predictions of the Efficient Market Hypothesis (Fama, 1970) and replicates the standard pattern documented in the FX literature, including Caulker (2020). These results justify modelling returns rather than price levels in the predictive regressions.

4.2 Autocorrelation and AR(1)

The AR(1) estimates for all four timeframes show that lagged returns contain almost no predictive information. The coefficients are extremely small in magnitude and statistically insignificant, and the Durbin–Watson statistics are all very close to 2, indicating the absence of autocorrelation. For example, the H4 estimate is $\hat{\phi} = 0.0020$ with $p = 0.927$, and $DW = 1.997$, implying that H4 returns behave as an essentially serially uncorrelated process. Similar results hold for H1, D1, and W1.

This pattern is consistent with Fama (1970), who notes that while small autocorrelations may occasionally appear in empirical work, they are typically too small to yield meaningful predictive content. The CHF/JPY series examined here fits this description closely.

4.3 EMA-50 Predictive Power

I evaluate the predictive value of the EMA–50 indicator using two forms of the signal: (i) a binary trend dummy, and (ii) the continuous price–EMA gap. In both cases, the regression results show no statistically significant relationship between today's EMA–based signal and next-period returns.

For the H4 timeframe, often believed by traders to exhibit the strongest trend structure, the EMA–50 gap coefficient is small and statistically insignificant ($\hat{\beta}_1 = -0.000054$, $p = 0.19$), with an R^2 value of only 0.0009. The trend-dummy specification performs similarly, with $\hat{\beta}_1 = -0.0000571$ and $p = 0.548$. Results for H1, D1, and W1 are likewise insignificant, with coefficient magnitudes near zero and negligible explanatory power.

Overall, the empirical evidence indicates that the EMA–50 does not possess statistically significant predictive power for CHF/JPY returns on any of the four timeframes examined. While the indicator is widely used by practitioners as a trend filter, its signals do not translate into measurable forecasting ability in this dataset. This finding is consistent with the weak-form EMH, which posits that publicly available price information should not enable systematic excess returns.

Chapter 5

Comparison with Previous Research

5.1 Relation to Fama (1970)

My results align closely with the weak-form Efficient Market Hypothesis as articulated by Fama (1970). The CHF/JPY price series behaves as a non-stationary random walk, and the return series exhibit extremely small and statistically insignificant autocorrelations. This fits Fama's observation that, although minor serial dependencies may appear in some empirical studies, they are typically too small to provide exploitable forecasting power (pp. 393–394).

Importantly, the EMA–50 indicator, despite its widespread use among traders, does not produce statistically significant predictability in my regression tests. This supports Fama's conclusion that liquid financial markets may display "almost but not perfectly" random behavior, yet still remain too efficient for simple technical rules to generate systematic excess returns.

5.2 Relation to Caulker (2020)

Caulker's (2020) analysis of USD/EUR documents volatility clustering, departures from perfect efficiency, and profitable short-term trading strategies based on volatility forecasts. My study is related in spirit but examines a different question, whether a simple trend-following indicator such as the EMA–50 can forecast *directional* returns in the CHF/JPY market.

In contrast to Caulker's findings, my results show no statistically significant predictive power in the EMA–50 signal across any timeframe. This difference highlights the importance of distinguishing between models that predict *volatility* and those that attempt to predict *returns*. While volatility is often forecastable in FX markets, directional return predictability remains extremely limited, consistent with weak-form EMH.

5.3 Relation to Earlier Work

My earlier project on technical analysis in forex markets relied primarily on visual inspection and cumulative return simulations, which suggested that EMA-based strategies aligned well with observed price movements. The present thesis extends that work by conducting formal econometric tests of return predictability.

Specifically, I test for unit roots, examine autocorrelation structures, and estimate predictive regressions to evaluate whether EMA–50 contains statistically significant information about future returns. The findings show that, once returns are modeled appropriately, EMA–50 signals do not provide predictive power, reinforcing the importance of rigorous statistical testing when evaluating technical indicators.

Chapter 6

Discussion

6.1 Interpretation of the H4 Results

Although traders commonly regard the H4 timeframe as exhibiting clearer trend structure than the H1, D1, or W1 frequencies, the empirical evidence in this thesis shows that the EMA–50 indicator does not possess statistically significant predictive power even at this intermediate horizon. The AR(1) coefficient for H4 returns is extremely small ($\hat{\phi} = 0.002$) and statistically insignificant, and the EMA–based regression coefficients are likewise close to zero with negligible explanatory power.

These findings suggest that any apparent trends observed visually on H4 charts do not translate into measurable forecasting ability once subjected to formal econometric tests. The visual smoothness of H4 price movements may create the impression of predictable structure, but the underlying return process remains close to serially uncorrelated noise.

6.2 Implications for EMH

The results are strongly consistent with the weak-form Efficient Market Hypothesis (EMH). Price levels behave as non-stationary random walks, return series are stationary, and lagged information, including EMA–50 signals does not help forecast future returns in a statistically meaningful way. This indicates that publicly available price information is rapidly incorporated into CHF/JPY exchange rates, leaving little room for simple technical indicators to generate systematic excess returns.

These findings align with Fama (1970)'s nuanced interpretation of EMH: markets may display minor deviations from perfect randomness, yet such deviations are typically too small to be exploited profitably. While traders often rely on moving averages as descriptive tools for identifying perceived trends or regimes, the econometric evidence presented here suggests that these signals do not contain predictive information about future returns in the CHF/JPY market.

6.3 Broader Interpretation

The absence of predictive power in EMA–50 also highlights an important distinction between visual interpretability and statistical predictability. Moving averages smooth noisy price series and make trend patterns easier to perceive, which may explain their widespread use among practitioners. However, smoothing alone does not create forecastable structure. The results of this thesis support the view that technical indicators may be useful for *describing* market behavior but not necessarily for *predicting* it.

Moreover, the contrast between trader intuition and statistical outcomes reflects a broader insight in behavioral finance: humans tend to detect patterns even in largely random data. EMA–based strategies may therefore persist in practice due to psychological appeal, ease of interpretation, and alignment with common trading heuristics, even when their predictive content is negligible.

Chapter 7

Conclusion

This thesis examined whether the 50-period Exponential Moving Average (EMA–50) contains predictive power for CHF/JPY returns and how the findings relate to the Efficient Market Hypothesis (EMH). Using a step-by-step econometric approach, including stationarity tests, autocorrelation analysis, AR(1) estimation, Durbin–Watson diagnostics, and predictive regressions, I arrive at the following conclusions:

1. CHF/JPY prices behave as non-stationary random walks, while log returns are stationary. This pattern is consistent with standard financial theory and prior FX studies.

2. Autocorrelation analysis and AR(1) estimation show that CHF/JPY returns exhibit extremely small and statistically insignificant serial dependence across all timeframes (H1, H4, D1, W1). Lagged returns therefore contain almost no predictive information.

3. Both EMA–50 measures tested,the binary trend dummy and the continuous price,EMA gap fail to produce statistically significant coefficients in predictive regressions. The associated R^2 values are effectively zero, indicating that EMA–50 signals do not explain variation in next-period returns.

4. These results support the weak-form EMH: publicly available price information is rapidly incorporated into CHF/JPY exchange rates, leaving little room for simple technical indicators to generate systematic excess returns.

5. The findings differ from studies that report short-term predictability or profitable technical strategies, such as Caulker (2020), highlighting the importance of distinguishing between volatility forecasting and directional return predictability.

Overall, the empirical evidence indicates that the EMA–50, despite its widespread use among traders as a trend filter, does not possess predictive content for CHF/JPY returns in a statistically meaningful sense. For practitioners, EMA–50 may remain a useful descriptive tool for visualizing market structure, but it should not be relied upon for systematic forecasting.

For researchers, these results reinforce the robustness of the weak-form EMH in highly liquid FX markets underscore the importance of rigorous econometric evaluation when assessing the effectiveness of technical trading rules.

References

Bibliography

Fama, E. F. (1970). *Efficient Capital Markets: A Review of Theory and Empirical Work*. Journal of Finance, 25(2), 383–417.

Caulker, S. (2020). *Short-Term Trading Strategy of Foreign Currency in an Emerging Market*. SSRN Electronic Journal.

Brock, W., Lakonishok, J., & LeBaron, B. (1992). *Simple Technical Trading Rules and the Stochastic Properties of Stock Returns*. Journal of Finance.

Lo, A., Mamaysky, H., & Wang, J. (2000). *Foundations of Technical Analysis*. Journal of Finance.

Campbell, J. Y., Lo, A. W., & MacKinlay, A. C. (1997). *The Econometrics of Financial Markets*. Princeton University Press.

Sullivan, R., Timmermann, A., & White, H. (1999). *Data-Snooping, Technical Trading Rule Performance, and the Bootstrap*. Journal of Finance, 54(5), 1647–1691.